

Aidan Burrowes

aidan.burrowes@gmail.com | (407) 925-6063 | [linkedin.com/in/aidanburrowes](https://www.linkedin.com/in/aidanburrowes) | [aidanburrowes.github.io](https://github.com/aidanburrowes)

Education

University of Florida, Gainesville, FL

May 2026

Bachelor of Science in Computer Science, Minors in Electrical Engineering & Physics

GPA 3.9/4.0, UF Honors Program

Coursework: Operating Systems, Networks, Databases, Intro to ML, Data Structures, Algorithms, Digital Logic, Intro to VR, Game Engine Dev, Linear Algebra, Quantum Computing, Systems Programming, Signals & Systems

Activities: Roblox – Campus Ambassador, Handshake AI – QA Tester, Perplexity – Campus Partner, Florida Quantum Computing Society – VP, ColorStack – Mentor & Project Lead, NSBE – Mentor, SHPE – Member

Work Experience

Roblox, San Mateo, CA

May 2025 – Aug 2025

Software Engineer Intern

- Developed novel CI break detection algorithm, **increasing CI efficiency by 50%, saving 15hr/wk & \$50k/yr**
- Created dynamic, automated TeamCity configs. for CI builds, **reducing developer life cycle from 24h to 5m**
- Built full stack app & microservice architecture—React, TypeScript, Go, Docker, AWS, Redis, GHA, HashiStack; **improved productivity by ~15% across 250+ engineers** via real-time build analytics & new query language

Microsoft, Redmond, WA

May 2024 – Aug 2024

Software Engineering Intern

- Enhanced Python CLI tool, using LLMs/TTS to create synthetic meeting audio & transcripts for 150 languages
- Optimized tool efficiency & consistency, scaling audio output by 400% & reduced hallucinations to under 2%
- Created CI/CD pipelines via Azure DevOps, **generating 20 datasets** to expedite GPT-4o fine-tuning & testing
- Fine-tuned STT models, **increasing entity transcription accuracy by 50%** in Teams meetings for **1m+ users**

HumanAI, Singapore, Singapore

May 2023 – Jul 2023

Software Engineer Intern

- Built full stack mobile app via Flutter, Node.js, Express.js, & MongoDB, enabling users to control robotic chef
- Optimized data pipeline via caching & lazy loading, **achieving 100% faster application loading times**
- Engineered ingredient-drop mechanism via CAD & Raspberry Pi, reducing dispensing latency from 15s to 5s
- Redesigned UI in Figma and built Swift/SwiftUI scaffold, **reducing onboarding time for interns by 2 weeks**

Research Experience & Projects

Data-Driven Humanities Research Group, Gainesville, FL

Jan 2023 – May 2026

Undergraduate Research Assistant

- Earned competitive AI Scholar designation (1 of 35) & **\$1,750 funding** to lead team of 5 in Latin NLP research
- Processed 10M+ tokens via Python & PostgreSQL, **expanding dataset by 50%** to improve model performance
- Fine-tuned LLMs on Latin syntax via TensorFlow, achieving up to **98% accuracy** on classification tasks
- Led team of 8 to build ETL & MLOps pipelines, reducing model deployment time from **12 hours to 1 hour**

UF Intelligent Clinical Care Center, Gainesville, FL

Jun 2025 – Dec 2025

Machine Learning Engineer Intern

- Created scalable AI agent infrastructure for task automation via MCP, LangChain, FastAPI, Docker, & RAG
- Automated 5 HR & 8 engineering workflows, **reducing human work by 10 hr/wk & saving \$15k/yr**
- Designed modular Msty configurations, enabling **20+ users to create pipelines** w/o engineering support

Gator Glaciology, Gainesville, FL

Aug 2023 – May 2024

Undergraduate Research Assistant

- Optimized GStatSim, a Python geostatistics package, using PyTorch & GPU acceleration on HPC clusters
- Expedited interpolation of Antarctic/Greenlandic bed topography maps, achieving a **4x speedup (8m to 2m)**

Publications & Professional Service

- Coauthor**, 2D spintronics for neuromorphic computing, *JLPEA*, doi.org/10.3390/jlpea15020016 Mar 2025
- Coauthor**, sentiment analysis for ancient Greek literature, *DSH*, doi.org/10.1093/llc/fqae048 Sep 2024
- Author**, syntactic analysis of ancient Latin literature using AI, *UF JUR*, [\[Paper presentation\]](#) May 2024
- Reviewer**, Oxford University Press, *Digital Scholarship in the Humanities* Jan 2025 – May 2026

Skills

Languages | Python, Swift, Java, Golang, C++, TypeScript, JavaScript, SQL, MATLAB, Dart, HTML/CSS, Assembly

Technologies | PyTorch, TensorFlow, AWS, GCP, Docker, Nomad, HashiStack, GraphQL, Webpack, Kibana, Spark MongoDB, PostgreSQL, CRDB, Jenkins, Kubernetes, Kafka, FastAPI, Flask, gRPC, React, Node.js, Express.js, Figma, Unix/Linux, macOS, Git, Unity, Android Studio, SwiftUI, NVIDIA Isaac Sim, Cursor, Claude Code, Codex, OpenClaw